

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech

Branch : Computer Engineering/Computer Science & Engineering

Subject Code & Name: BTCOE703C Big Data Analytics

Semester : VII

Max. Marks: 60

Date: 10/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. Objective type questions. (Compulsory Question)			12
1			
1	Transaction of data of the bank is a type of -----.	L1/CO1	1
	a. Unstructure d. data b. Structured data c. Social data d. Machine Data		
2	_____ is the slave/worker node and holds the user data in the form of Data Blocks.	L1/CO3	1
	a. DataNode b. NameNode c. Data block d. Replication		
3	The _____ is responsible for allocating resources to the various running applications subject to familiar constraints of capacities, queues.	L2/CO2	1
	a. Manager b. Master c. Tracker d. Scheduler		
4	How much will bytes for 1Petabytes	L3/CO1	1
	a. 1024bytes b. 1024 ⁵ bytes c. 1024 ⁴ bytes d. 1024 ⁶ bytes		
5	In case of Yet Another Resource Negotiator (YARN), the _____ User Interface provides host and port information.	L2/CO2	1
	a. Data Node b. Name Node c. Replication d. Resource Manager		
6	Kafka is comparable to traditional messaging systems such as _____	L2/CO3	1
	a. Impala b. Zookeeper c. ActiveMQ d. BigTop		
7	Mapper implementations are passed the JobConf for the job via the _____ method.	L3/CO2	1
	a. JobConfigure .configure b. JobConfigurable. configurable c. JobConfigurabl e.configure d. JobMapConfig ure.configure		
8	Mahout provides an implementation of a _____ identification	L2/CO4	1

	algorithm which scores collocations using log-likelihood ratio.		
	a.collocation b.compaction c.collection d. Corelation		
9	What is mean by "collection" in MongoDB?	L2/CO5	1
	a.A group of tables b. A group of documents c. A single document d. A type of index		
10	_____ leverages Spark Core fast scheduling capability to perform streaming analytics.	L2/CO3	1
	a.Mllib b. GraphX c. Index d. Spark Streaming		
11	Which of the following format is supported by MongoDB?	L2/CO5	1
	a.BSON b. XML c.SQL d. NoSQL		
12	_____ is a distributed graph processing framework on top of Spark.	L2/CO4	1
	a. Spark b. ML library c.GraphX d. BSON		
Q. 2	Solve the following.		12
A)	What is a big data? Discuss the different sources of big data.	L2/CO1	6
B)	Describe the various characteristics of Big data.	L2/CO1	6
Q.3	Solve the following.		12
A)	With neat Sketch explain HDFS architecture in detail.	L2/CO2	6
B)	Explain the Map Reduce programming model. Solve the example of Word Count for 1.This is an Apple. 2.An Apple is red in colour.	L3/CO2	6
Q. 4	Solve Any Two of the following.		12
A)	Why is Real-time Big Data Pipeline So Important Nowadays?	L2/CO3	6
B)	Explain the Role and working of APACHE KAFKA.	L2/CO3	6
C)	Discuss the typical use cases of streaming data.	L2/CO3	6
Q.5	Solve Any Two of the following.		12
A)	What is Machine Learning? Why the Machine Learning is important in Big Data processing?	L2/CO4	6
B)	What is Graph processing? Discuss the Graph processing algorithms in big data.	L2/CO4	6
C)	Explain the K-Means clustering Algorithm with example.	L2/CO4	6
Q. 6	Solve Any Two of the following.		12
A)	Discuss the advantages of using Java script shell for MongoDB queries	L3/CO5	6

B)	Explain the CRUD operations in MongoDB with example.	L2/CO5	6
C)	What is the NoSQL database? Explain different types of NoSQL databases.	L2/CO5	6
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